

UI/UX Design Document

Invoice Processing — From PDF to Decision | Implementation-Ready UX

1. Design Goal

The UI should make the system feel like an AP analyst's control room, not an AI chatbot. The most important question on every screen is: What happened, what did the system compare, and why is this safe or unsafe to approve?

2. Navigation

Route	Purpose
/	Dashboard / run history
/process	Upload and start a run
/runs/{id}	Live processing + decision detail
/runs/{id}/review	Human review
/reference/pos	PO reference data
/reference/vendors	Vendor reference data

3. Primary User Journey

1. Open dashboard.
2. Click Process Invoice.
3. Drop PDF or browse.
4. Preview filename/page count and click Process.
5. Watch live stage timeline: Uploaded → Extracted → Vendor Resolved → PO Candidates → PO Match → Validated → Duplicate Check → Decision.
6. See decision banner: APPROVE / REVIEW / REJECT.
7. Inspect invoice fields and PO comparison.
8. If REVIEW, select/confirm a PO or record an override reason.
9. Return to dashboard history.

4. Screen: Dashboard

- Top action: Process Invoice.
- Summary cards: Total Runs, Approved, Review, Rejected, Average Processing Time.
- Recent runs table: time, vendor, invoice #, amount, PO, decision, confidence/status.
- Filters: decision, vendor, date, exception type.
- Each row opens the run detail.

5. Screen: Upload

- Large drag-and-drop area with PDF icon.
- Accepted: PDF; show maximum file size.
- After selection show file name, size, page count if available.
- Primary CTA: Process Invoice.
- Validation error appears inline and explains how to fix it.

6. Screen: Live Run

Component	Behavior
Run header	Run ID, vendor when known, elapsed time, current status
Stage timeline	Vertical list with pending/running/success/warning/error
Stage detail	Expandable JSON-like human-readable evidence, not raw model dump
Progress	Indeterminate or stage-based; never fake percentage
Decision panel	Appears once policy engine finishes
Cancel/retry	Retry only safe failed technical stages; never silently rerun a financial decision

7. Screen: Decision Detail

- Large status badge: APPROVE / REVIEW / REJECT.
- One-sentence reason at top.
- Invoice summary: vendor, invoice #, date, currency, subtotal, tax, total.
- PO summary: PO #, vendor, status, original value, remaining value.
- Side-by-side comparison table: invoice vs PO/remaining PO.
- Checks list: Vendor, PO status, currency, quantity, unit price, subtotal, tax, total, tolerance, duplicate.
- PO candidates section with score and reasons.
- Evidence section showing extraction confidence/source page.
- Audit timeline.

8. Review UX

Review should be an exception workspace, not a second dashboard.

- Show the reason for escalation first.
- Show only the relevant evidence and failed/uncertain checks prominently.
- If multiple POs are candidates, show candidate cards with score breakdown.
- Allow 'Select this PO' only to authorized users.
- Require a review note for any manual resolution.
- Show the final decision after submission and preserve the original automated decision.

9. Visual Language

Element	Guideline
Typography	Inter/Aptos-like sans serif; 14–16px body; strong numeric hierarchy
Spacing	8px base spacing system; generous card padding
Cards	Subtle border, low-radius or moderate radius; avoid excessive shadows
Decision status	Use distinct semantic status colors with text labels; never rely on color alone
Tables	Compact, sticky headers for comparison; right-align monetary values
Numbers	Use consistent currency formatting and preserve source values when relevant
Icons	Simple line icons; always paired with text for critical actions

10. Accessibility

- Keyboard navigable controls.
- Visible focus states.
- Semantic headings and labels.
- Status is communicated with text + icon, not color only.
- Tables have clear headers.
- Error messages are attached to the relevant input.
- Sufficient contrast for body text and controls.

11. Responsive Behavior

- Desktop is the primary AP workflow.
- At tablet width, side-by-side invoice/PO panels stack.
- On narrow screens, comparison tables become horizontally scrollable and decision summary remains pinned at top.
- Live timeline remains readable without requiring hover interactions.

12. UI States

State	Design
Empty	Explain what the page does + primary action
Loading	Show real stage currently running; avoid fake percentages
Success	Decision + evidence + next action
Warning	Amber/attention state with exact issue
Error	Technical failure with retry guidance; no decision shown
Review	Strong exception banner and actionable evidence
No PO match	Show 'No confident PO match' and the search evidence
Multiple matches	Show candidates and why the system could not safely choose

13. Key Interaction Details

- Clicking a failed validation opens the exact comparison and discrepancy.
- Clicking a PO candidate expands its signal breakdown.
- Hover is never required to understand a critical decision.
- Raw JSON is available in a collapsible 'Technical details' section for the demo, while the default view remains business-friendly.
- The final decision panel remains stable even when users inspect lower-level evidence.

14. Design Principles

10. Explain before exposing complexity.
11. Show evidence before confidence scores.
12. Never hide uncertainty.
13. Make exceptions actionable.
14. Keep the primary workflow to one obvious path.
15. Do not pretend that model output is a financial control.